

Week – Hypothesis testing II & A/B

Business Forecasting

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Where we are

C1 Framing → C2 Describing data → **C3 Uncertainty** → C4–C5 Regression → C6 Forecasting

We are in **C3 — uncertainty**. Last week we tested claims about **one** parameter. This week we **compare two groups** and design experiments.

- **This week (Testing II): two-sample & paired tests, correlation, and A/B testing** — the causal-inference tool behind modern business experiments.

Fast & hands-on: every test you run in R (`t.test` , `cor.test`) — practice in `W07_learnr.Rmd` .

Hypothesis Testing: Two samples

Two Samples and the Difference in Means

Two iid samples **independent** of each other from two populations $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$.

General idea: test the difference between **population means**.

- Ex: $H_0 : \mu_X - \mu_Y = 100$ vs $H_A : \mu_X - \mu_Y \neq 100$
- Ex: $H_0 : \mu_X - \mu_Y = 0$ vs $H_A : \mu_X - \mu_Y > 0$
- Test for no difference: $H_0 : \mu_X - \mu_Y = 0$ vs $H_A : \mu_X - \mu_Y \neq 0$
 - **Test statistic:** normalized difference in sample means

$$\text{test stat} = \frac{\bar{X} - \bar{Y} - \overbrace{(\mu_{X,0} - \mu_{Y,0})}^{\Delta_0}}{SE} = \frac{\bar{X} - \bar{Y} - \Delta_0}{SE}$$

Variance in two samples

- SE depends on the estimate of the standard deviation:
 - If standard deviation known: $SE = \sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$
- Why? Because $Var(\bar{X} - \bar{Y}) = Var(\bar{X}) + Var(\bar{Y}) = \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}$
- If standard deviation not known: $SE = \sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}$
- **Rejection region:** depends on the distribution
 - Large sample or known variance from normal: $Z_{test} \sim N(0, 1)$
 - Small sample with normal data: $T_{test} \sim t(v)$ — critical values from Student's t, v df

Side note: degrees of freedom for a difference in means

Two ways to do it:

1. Proper, complex (and annoying):

$$v = \frac{\left(\frac{s_X^2}{n} + \frac{s_Y^2}{m} \right)^2}{\frac{(s_X^2/n)^2}{n-1} + \frac{(s_Y^2/m)^2}{m-1}}$$

Round down to the nearest integer.

1. Simple, approximate shortcut:

$$v = \min(n - 1, m - 1)$$

Difference in means - Two Sided Test

Hypothesis: $H_0 : \mu_x - \mu_Y = \Delta_0$ and $H_A : \mu_x - \mu_Y \neq \Delta_0$

Rejection Region: for a test of significance level α , reject

- If large sample or known variance from normal

$$\frac{\bar{X} - \bar{Y} - \Delta_0}{SE} < z_{\frac{\alpha}{2}} \quad or \quad \frac{\bar{X} - \bar{Y} - \Delta_0}{SE} > z_{1-\frac{\alpha}{2}}$$

- If small sample from normal

$$\frac{\bar{X} - \bar{Y} - \Delta_0}{SE} < t_{v, \frac{\alpha}{2}} \quad or \quad \frac{\bar{X} - \bar{Y} - \Delta_0}{SE} > t_{v, 1-\frac{\alpha}{2}}$$

Where $z_{1-\frac{\alpha}{2}}$ and $t_{v, 1-\frac{\alpha}{2}}$ are $(1 - \frac{\alpha}{2})$ quantiles of the standard normal and Student's t with v df.

Difference in means - Two Sided Test

If CLT kicks in, the distributions are:

$$\bar{X} \sim N(\mu_X, \sigma_X/\sqrt{n}) \quad \text{and} \quad \bar{Y} \sim N(\mu_Y, \sigma_Y/\sqrt{m})$$

The difference of two independent normals:

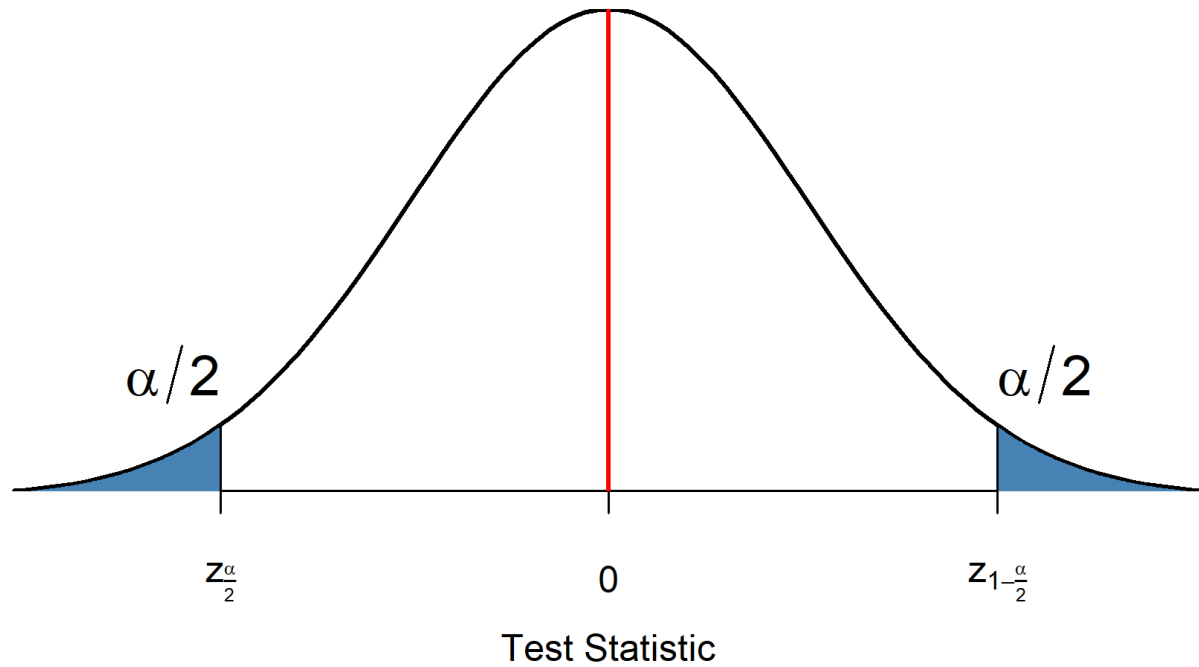
$$\bar{X} - \bar{Y} \sim N(\mu_X - \mu_Y, \sqrt{\sigma_X^2/n + \sigma_Y^2/m})$$

So if the null is true, the standardized statistic:

$$\frac{\bar{X} - \bar{Y} - (\mu_{X,0} - \mu_{Y,0})}{\sqrt{s_X^2/n + s_Y^2/m}} \sim N(0, 1)$$

► learnr W07_learnr.Rmd → Exercise 3: Decision at alpha = 0.05

Distribution under H0



Confidence interval for a difference

Following this idea, we can construct a confidence interval for the difference:

$$P\left(z_{\alpha/2} < \frac{\bar{X} - \bar{Y} - (\mu_{X,0} - \mu_{Y,0})}{\sqrt{s_X^2/n + s_Y^2/m}} < z_{1-\alpha/2}\right) = 1 - \alpha$$

$$CI_{1-\alpha} = \left(\bar{X} - \bar{Y} \pm z_{1-\alpha/2} \sqrt{s_X^2/n + s_Y^2/m}\right)$$

► `learnr W07_learnr.Rmd` → Exercise 2: P-value and Confidence Interval

Two Samples - Mean - One sided - Case 1

Hypothesis: $H_0 : \mu_X - \mu_Y = \Delta_0$ and $H_A : \mu_X - \mu_Y < \Delta_0$

- This includes any null like $\mu_X - \mu_Y \geq \Delta_0$. Why?
- If you rejected $\mu_X - \mu_Y = \Delta_0$ at α , you would reject for sure anything larger than Δ_0 . **Rejection Region:** for a test of significance level α , reject
- If large sample or known variance from normal

$$\frac{\bar{X} - \bar{Y} - (\Delta_0)}{SE} < z_\alpha$$

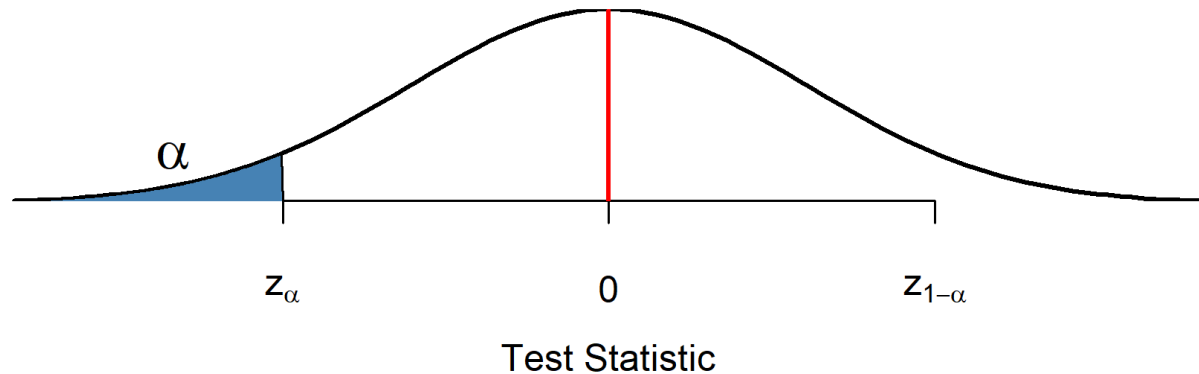
- If small sample from normal

$$\frac{\bar{X} - \bar{Y} - (\Delta_0)}{SE} < t_{v,\alpha}$$

Where z_α and $t_{v,\alpha}$ are α quantiles of the standard normal and Student's t with v df.

Two Samples - Mean - One sided - Case 1

Distribution under H0



- **Intuition:** never reject if $\bar{X} - \bar{Y} \geq \Delta_0$; reject if $\bar{X} - \bar{Y}$ is sufficiently smaller than Δ_0 . If the null is true we reject with probability

$$P\left(\frac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{s_X^2/n + s_Y^2/m}} < z_\alpha\right) = P(Z < z_\alpha) = \alpha$$

Two Samples - Mean - One sided - Case 2

Hypothesis: $H_0 : \mu_X - \mu_Y = \Delta_0$ and $H_A : \mu_X - \mu_Y > \Delta_0$

- This includes any null like $\mu_X - \mu_Y \leq \Delta_0$. Why?
- If you rejected $\mu_X - \mu_Y = \Delta_0$ at α , you would reject for sure anything smaller than Δ_0 . **Rejection Region:** for a test of significance level α , reject
- If large sample or known variance with normal

$$\frac{\bar{X} - \bar{Y} - (\Delta_0)}{SE} > z_{1-\alpha}$$

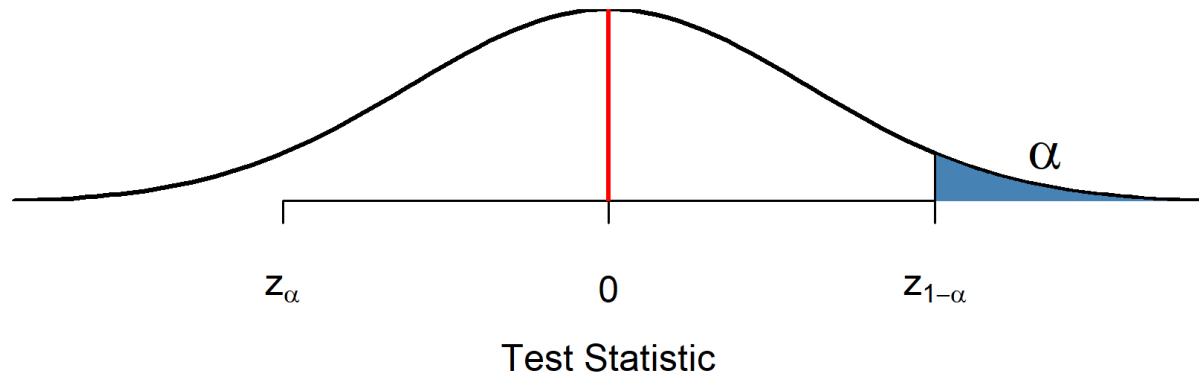
- If small sample from normal

$$\frac{\bar{X} - \bar{Y} - (\Delta_0)}{SE} > t_{v,1-\alpha}$$

Where $z_{1-\alpha}$ and $t_{v,1-\alpha}$ are $(1 - \alpha)$ quantiles of the standard normal and Student's t with v df.

Two Samples - Mean - One sided - Case 2

Distribution under H0

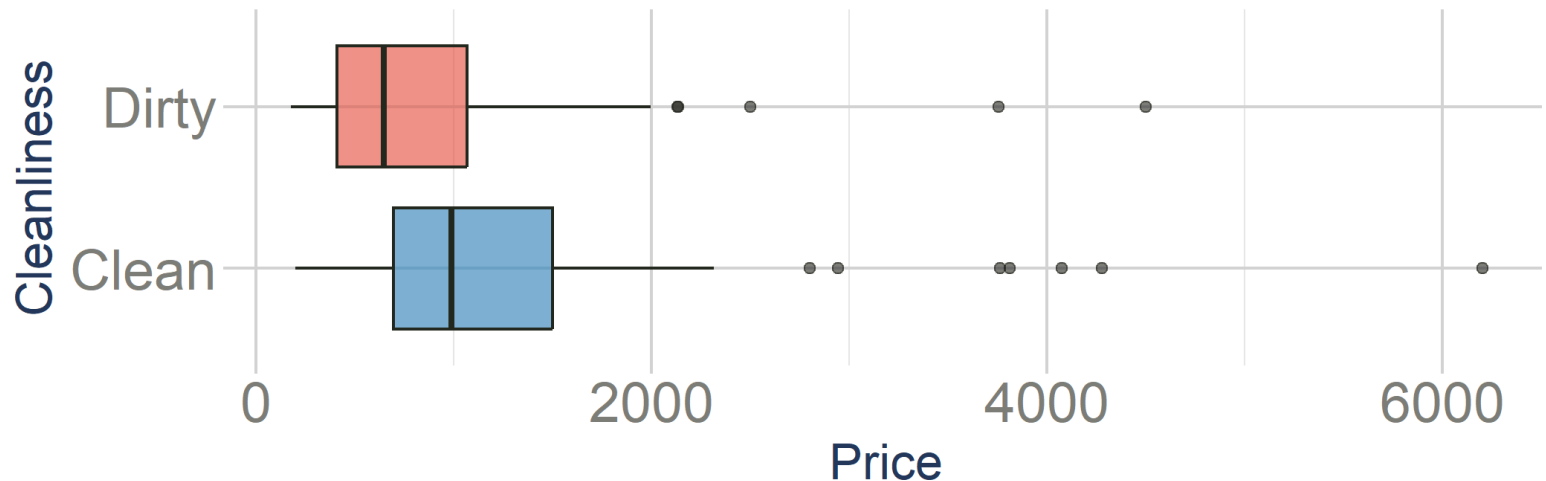


- **Intuition:** never reject if $\bar{X} - \bar{Y} \leq \Delta_0$; reject if $\bar{X} - \bar{Y}$ is sufficiently larger than Δ_0 . If the null is true we reject with probability

$$P\left(\frac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{s_X^2/n + s_Y^2/m}} > z_{1-\alpha}\right) = P(Z > z_{1-\alpha}) = \alpha$$

Airbnb: Clean vs Dirty Prices

Sample	n	Mean	SD
Clean	100	1245	962
Dirty	100	869	693



- State the null and alternative hypothesis
- Calculate the value of the test statistic
- Calculate the p-value; at which α do we reject? Build the 95% CI for the difference.

► learnr [W07_learnr.Rmd](#) → Exercise 1: Two-Sample Difference in Means

Exercise: McDonald's vs Burger King

We suspect Burger King has larger caloric content for kids' meals. Let's test it.

Sample	n	Sample Mean	Sample Standard Deviation
McDonald	9	475	230
Burger King	16	520	180

- State the null and alternative hypothesis
- Calculate the value of the test statistic
- Determine the appropriate number of degrees of freedom
- Calculate the p-value; at which α do we reject?

► [learnr W07_learnr.Rmd](#) → Exercise 4: One-Sided Two-Sample Test

A/B Testing: The Business Question



- **Airbnb** wants to know: *Do professional photos increase host revenue?*
- **Historical data:** listings with better photos show higher revenues.
- But... **Correlation $\neq$ Causation.** Better photos may be linked to other factors: more organized hosts, higher-quality listings, better service.
- A simple comparison only tells us about **association**, not causality.

The Business Decision

- Should Airbnb **invest in professional photography** for hosts?
- If photos aren't the true driver → the investment wastes money. We need a method to **isolate the effect of photos**.

Enter A/B Testing

Definition:

- A method to measure *causal* effects.
- Widely used in modern business, especially tech.

Examples:

- Netflix: which thumbnail gets more clicks?
- Amazon: does a new checkout button increase sales?
- Instagram: do longer videos increase engagement?

The Setup

We divide hosts into **two groups**:

1. **Control group**: no intervention (existing photos).
2. **Treatment group**: Airbnb provides **professional photography**.

The KEY: groups are assigned **randomly**.

Why Randomization Matters

- Without randomization: groups differ in many ways.
- With randomization: groups are (on average) identical, except for photos.

This is what gives us **causal inference**!

► `learnr W07_learnr.Rmd` → Exercise 6: A/B Test as a Proportion Test

Classroom Exercise: Randomization Check

- Take a sample of Airbnb listings
- Randomly assign to two groups
- Compare characteristics (beds, location)

They should look very similar!

Why? We are taking two samples from the same population.

- All properties and expectations will be the same for both samples.

► `learnr W07_learnr.Rmd` → Exercise 8: What Makes an A/B Test Valid

Measuring the Effect

- After randomization: **the only difference is the photos.**
- Provide photography to the treatment group
- Wait to see potential effects
- Collect outcome data (revenue, reviews, bookings)
- Run a simple two-sample test between treatment and control
- If the treatment group earns more revenue, we know it was **caused by the professional photos**

This is the causal effect.

► `learnr W07_learnr.Rmd` → Exercise 7: Reading the Proportion Test

- A/B testing is used extensively in modern business, especially in tech.
- We have all been subjects of these experiments, often without knowing.

EXAM QUESTION — Fall 2025 Midterm 1

Q16. Design an **A/B test** to measure whether a change causes an increase in an outcome: define the treatment and control groups, the randomization, the outcome, and the two-sample test you would run.

We frame this in class using `Data/households_environment.rda` (object `exam_data`):

- **Randomize** units into treatment vs control → the groups are, on average, identical except for the treatment.
- Collect the **outcome**, then run a **two-sample difference-in-means test** ($H_0 : \mu_T - \mu_C = 0$).

Practice now: the two-sample and A/B-design problems in the Week (Testing II) set.

Hypothesis Testing: Paired Data

Paired Data

- Sometimes we have only one set of individuals or objects, but two (or more) observations per each.
- **Example:** employee productivity — two observations per employee: productivity working from home and working from the office.
- **Example:** sales at Walmart and Soriana in each neighborhood — two observations per neighborhood.

Paired Data

We have a set of independent pairs of data $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$. Observations within a pair are not necessarily independent!

- Some productive workers are productive both at home and in the office; some unproductive are unproductive in both.
- We observe the difference for each individual: $X_i - Y_i = d_i$.
- We look at the population value of that difference: $\mu_X - \mu_Y = \Delta$.
 - Check if the means are the same: $\Delta = 0$ (no difference in productivity)
 - Or mean X is bigger than mean Y : $\Delta > 0$

Paired Data

Null hypothesis: $H_0 : \Delta = \Delta_0$

Test Statistic is:

$$\text{test statistic} = \frac{\overbrace{\bar{X} - \bar{Y}}^{\bar{d}} - \Delta_0}{SE} = \frac{\bar{d} - \Delta_0}{SE}$$

- Note on the standard error — it's not the same formula as for two independent samples:
 $\text{Var}(X_i - Y_i) \neq \text{Var}(X_i) + \text{Var}(Y_i)$
- We just calculate it from the differences:
 - If we know the variance, $SE = \frac{\sqrt{\text{var}(d_i)}}{\sqrt{n}} = \frac{\sigma_d}{\sqrt{n}}$
 - If we don't: $SE = \frac{s_d}{\sqrt{n}}$

Paired Data

Rejection Region: for a test of significance level α :

- If large sample or known variance with normal distribution:
- for $H_A : \bar{d} \neq \Delta_0$, reject if:

$$\frac{\bar{d} - \Delta_0}{SE} < z_{\frac{\alpha}{2}} \quad \text{or} \quad \frac{\bar{d} - \Delta_0}{SE} > z_{1-\frac{\alpha}{2}}$$

- for $H_A : \bar{d} > \Delta_0$, reject if:

$$\frac{\bar{d} - \Delta_0}{SE} > z_{1-\alpha}$$

- for $H_A : \bar{d} < \Delta_0$, reject if:

$$\frac{\bar{d} - \Delta_0}{SE} < z_{\alpha}$$

- If small sample from normal, replace critical values with the [t-distribution with n-1 degrees of freedom](#).

Paired or Unpaired Data?

1. We want to know if Intel's stock and Southwest Airlines' stock have similar rates of return. We take a random sample of 50 days and record both stocks on those same days.
 - Paired
1. We randomly sample 50 items at Target and note each price. Then we visit Walmart and collect the price for each of those same 50 items.
 - Paired
1. A school board wants to know if average SAT scores differ between two high schools. They take a simple random sample of 100 students from each school.
 - Not paired

► `learnr W07_learnr.Rmd` → Exercise 5: Paired or Unpaired?

Exercise: Does IQ Decrease with Age?

A psychologist thinks age influences IQ. They take a random sample of 100 people aged 40. For each person we know their IQ at age 16 and now. On average, IQ at the young age was 8 points higher than at age 40, with a standard deviation of that difference of 7 points. Using $\alpha = 0.01$, test the hypothesis that IQ decreases with age.

LearnR: Paired Data test for means — work through this IQ example in R

Hypothesis Testing for Correlation

Example: are cleanliness score (C) and price (P) correlated in the population?

Assume **normally distributed variables** in **independent pairs** $\{(C_1, P_1), (C_2, P_2), \dots\}$.

Null hypothesis: $H_0 : \rho(C, P) = \rho_0$

Test statistic and its distribution under the null:

$$T_{test} = \frac{\hat{\rho}(C, P) - \rho_0}{\sqrt{\frac{1 - \hat{\rho}(C, P)^2}{n-2}}} \sim t_{n-2}$$

where $\hat{\rho}(C, P)$ is the sample correlation coefficient.

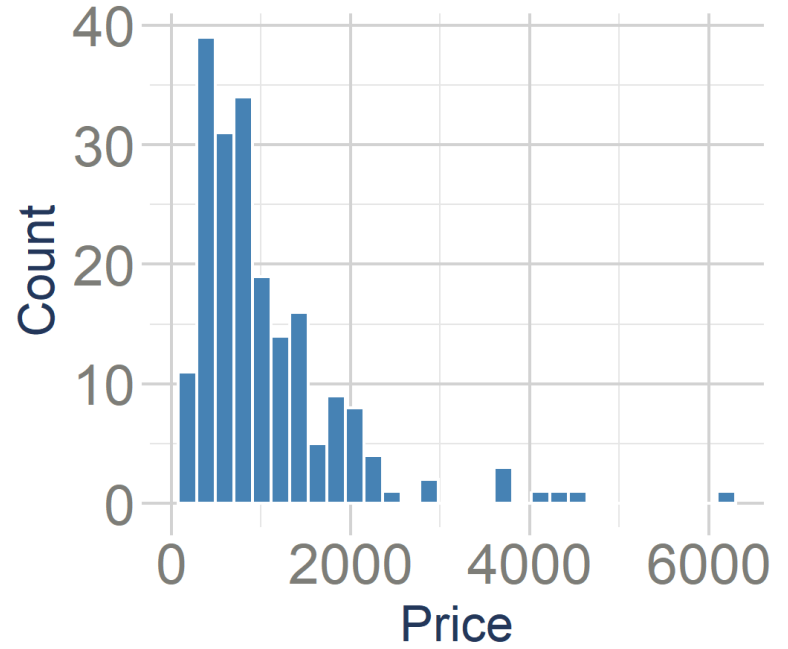
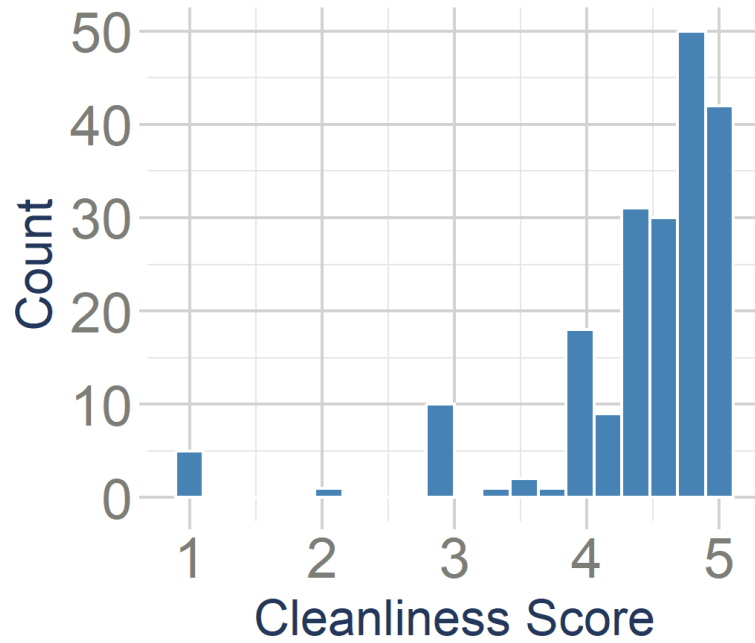
Alternatives and their **rejection regions**:

- $H_A : \rho \neq \rho_0$ — reject if $t_{test} > t_{1-\frac{\alpha}{2}, n-2}$ or $t_{test} < t_{\frac{\alpha}{2}, n-2}$
- $H_A : \rho > \rho_0$ — reject if $t_{test} > t_{1-\alpha, n-2}$
- $H_A : \rho < \rho_0$ — reject if $t_{test} < t_{\alpha, n-2}$

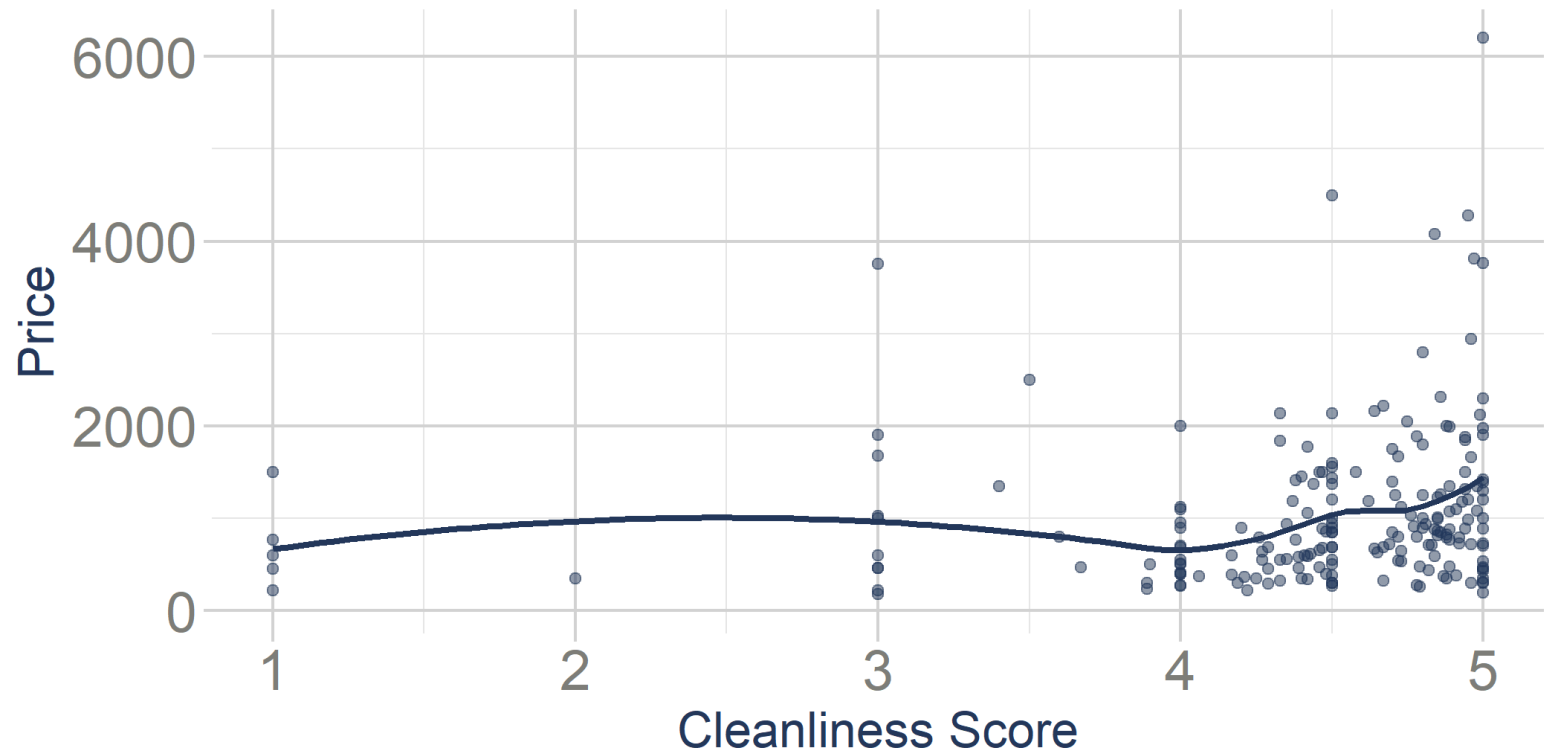
Correlation test

- Suppose we test $H_0 : \rho(C, P) = 0$ vs $\rho(C, P) > 0$ in our Airbnb data (clean score vs price).
- We **assume**: variables are normally distributed; pairs are independent; the relationship is linear.
- Sample correlation is 0.159, with $n = 200$.
- So $t_{test} = \frac{\hat{\rho}\sqrt{(n-2)}}{\sqrt{1-\hat{\rho}^2}} = \frac{0.159\sqrt{198}}{\sqrt{1-0.025}} = 2.27$
- Since $n = 200$, Student's t is identical to the standard normal. Either compare to critical regions, or compute the p-value:
- $p\text{-value} = P(Z > t_{test}) = P(Z > 2.27) = 0.0126$
 - A test at $\alpha = 0.05$ would reject H_0 .

Correlation Test – Distributions



Correlation Test — Scatter Plot



Exercise: Correlation Test

Suppose you have a random sample of 27 units (from a bivariate normal). X measures a client's age and Y their spending. You calculated the correlation coefficient $\hat{\rho} = -0.45$. Can you reject the null $\rho = 0$ in favor of the alternative $\rho \neq 0$ at the 5% significance level?

LearnR: Paired Data Test for Correlation — compute the t-statistic for this exercise

What you must be able to do this week

- **Two-sample difference in means:** build the test statistic, choose the SE and degrees of freedom, and reach a decision; construct a CI for the difference.
- **Paired data:** recognize when data are paired, reduce to differences d_i , and run the one-sample test on them.
- **Correlation test:** compute $t_{test} = \hat{\rho} \sqrt{\frac{n-2}{1-\hat{\rho}^2}}$ and test $H_0 : \rho = 0$.
- **A/B testing:** design a randomized experiment, explain why randomization delivers **causal** inference, and run the two-sample test on the outcome.

Course arc: with testing done, C4–C5 move from comparing groups to **modeling** relationships — regression.